

LAB 1 — USE CASE FLOW

UC-04: Submit Emergency Blood Request

BLOOD BANK DISCOVERY AND EMERGENCY DONOR

Primary Actor:

Emergency Requester

Supporting Actors:

Blood Bank Manager, SMS Notification Service, Eligible Donor

Goal:

To submit an emergency blood request, check whether the required blood is available, and find suitable nearby donors if needed.

Main Success Scenario:

1. The Emergency Requester selects "Submit Emergency Blood Request".
2. The system shows the emergency blood request form.
3. The Emergency Requester enters the required blood group, number of units needed, and emergency location.
4. The system checks whether the entered details are valid.
5. The system creates the emergency request and generates a unique request ID.

6. The system checks the current blood inventory for the requested blood group.
7. The system shows the available number of units and checks whether the required quantity is available.
8. If enough blood is not available, the system searches for compatible donors within 10 km of the emergency location.
9. The system displays the available matching donors.
10. The Emergency Requester or Blood Bank Manager selects a suitable donor to be notified.
11. The system sends an emergency SMS to the selected donor through the SMS Notification Service.
12. The system records the notification status and shows a confirmation message.
13. The use case ends successfully.

Alternate Flow — Insufficient Blood / No Donor:

Preconditions:

1. The Emergency Requester is authorized to submit an emergency blood request.
2. Blood inventory and donor information are available in the system.
3. The donor matching service is available.

Postconditions:

1. The emergency blood request is stored with a unique request ID.
 2. The blood inventory status is recorded for the request.
 3. If a suitable donor is found, the donor notification status is recorded.
 4. If no suitable donor is found, the emergency request remains active.
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- 7a. If the required blood units are not available, the system starts the emergency donor matching process.
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- 8a. If suitable donors are found within 10 km, the system displays the matching donors and continues from Step 10.
 - 8b. If no suitable donor is found, the system displays a message saying "No eligible donor found" and keeps the emergency request active for further action.

UML Relationships:

UC-04 Submit Emergency Blood Request «include» UC-05 Match Compatible Donors

UC-06 Send Emergency SMS Alert «extend» UC-05 Match Compatible Donors when there is a critical shortage.